

IS42/45R86400D/16320D/32160D IS42/45S86400D/16320D/32160D



**16Mx32, 32Mx16, 64Mx8
512Mb SDRAM**

**PRELIMINARY INFORMATION
JUNE 2011**

FEATURES

- Clock frequency: 200, 166, 143 MHz
- Fully synchronous; all signals referenced to a positive clock edge
- Internal bank for hiding row access/precharge
- Power supply: $V_{DD}/V_{DDQ} = 2.3V-3.6V$
IS42/45SxxxxxD - $V_{DD}/V_{DDQ} = 3.3V$
IS42/45RxxxxxD - $V_{DD}/V_{DDQ} = 2.5$
- LVTTTL interface
- Programmable burst length
– (1, 2, 4, 8, full page)
- Programmable burst sequence:
Sequential/Interleave
- Auto Refresh (CBR)
- Self Refresh
- 8K refresh cycles every 64 ms
- Random column address every clock cycle
- Programmable CAS latency (2, 3 clocks)
- Burst read/write and burst read/single write operations capability
- Burst termination by burst stop and precharge command
- Packages:
x8/x16: 54-pin TF-TSOP-II, 54-ball TF-BGA (x16 only)
x32: 90-ball TF-BGA
- Temperature Range:
Commercial (0°C to +70°C)
Industrial (-40°C to +85°C)
Automotive, A1 (-40°C to +85°C)
Automotive, A2 (-40°C to +105°C)

DEVICE OVERVIEW

ISSI's 512Mb Synchronous DRAM achieves high-speed data transfer using pipeline architecture. All inputs and outputs signals refer to the rising edge of the clock input. The 512Mb SDRAM is organized as follows.

PACKAGE INFORMATION

IS42/45S32160D	IS42/45S16320D	IS42/45S86400D
IS42/45R32160D	IS42/45R16320D	IS42/45R86400D
4M x 32 x 4 banks	8M x 16 x 4 banks	16M x 8 x 4 banks
90-ball TF-BGA	54-pin TSOP-II 54-ball TF-BGA	54-pin TSOP-II

KEY TIMING PARAMETERS

Parameter	-5	-6	-7	Unit
Clk Cycle Time				
CAS Latency = 3	5	6	7	ns
CAS Latency = 2	10	10	7.5	ns
Clk Frequency				
CAS Latency = 3	200	167	143	Mhz
CAS Latency = 2	100	100	133	Mhz
Access Time from Clock				
CAS Latency = 3	5.0	5.4	5.4	ns
CAS Latency = 2	6	6	5.4	ns

ADDRESS TABLE

Parameter	16M x 32	32M x 16	64M x 8
Configuration	4M x 32 x 4 banks	8M x 16 x 4 banks	16M x 8 x 4 banks
Bank Address Pins	BA0, BA1	BA0, BA1	BA0, BA1
Autoprecharge Pins	A10/AP	A10/AP	A10/AP
Row Address	8K(A0 – A12)	8K(A0 – A12)	8K(A0 – A12)
Column Address	512(A0 – A8)	1K(A0 – A9)	2K(A0 – A9, A11)
Refresh Count			
Com./Ind./A1	8K / 64ms	8K / 64ms	8K / 64ms
A2	8K / 16ms	8K / 16ms	8K / 16ms

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- a.) the risk of injury or damage has been minimized;
- b.) the user assume all such risks; and
- c.) potential liability of Integrated Silicon Solution, Inc is adequately protected under the circumstances

PIN FUNCTIONS

Symbol	Type	Function (In Detail)
A0-A12	Input Pin	Address Inputs: A0-A12 are sampled during the ACTIVE command (row-address A0-A12) and READ/WRITE command (column address A0-A9, A11 (x8); A0-A9 (x16); A0-A8 (x32); with A10 defining auto precharge) to select one location out of the memory array in the respective bank. A10 is sampled during a PRECHARGE command to determine if all banks are to be precharged (A10 HIGH) or bank selected by BA0, BA1 (LOW). The address inputs also provide the op-code during a LOAD MODE REGISTER command.
BA0, BA1	Input Pin	Bank Select Address: BA0 and BA1 defines which bank the ACTIVE, READ, WRITE or PRECHARGE command is being applied.
$\overline{\text{CAS}}$	Input Pin	$\overline{\text{CAS}}$, in conjunction with the $\overline{\text{RAS}}$ and $\overline{\text{WE}}$, forms the device command. See the "Command Truth Table" for details on device commands.
CKE	Input Pin	The CKE input determines whether the CLK input is enabled. The next rising edge of the CLK signal will be valid when is CKE HIGH and invalid when LOW. When CKE is LOW, the device will be in either power-down mode, clock suspend mode, or self refresh mode. CKE is an asynchronous input.
CLK	Input Pin	CLK is the master clock input for this device. Except for CKE, all inputs to this device are acquired in synchronization with the rising edge of this pin.
$\overline{\text{CS}}$	Input Pin	The $\overline{\text{CS}}$ input determines whether command input is enabled within the device. Command input is enabled when $\overline{\text{CS}}$ is LOW, and disabled with $\overline{\text{CS}}$ is HIGH. The device remains in the previous state when $\overline{\text{CS}}$ is HIGH.
DQM: x8 DQML, DQMH: x16 DQM0-DQM3: x32	Input Pin	DQx pins control the bytes of the I/O buffers. For example with x16, in read mode, DQML and DQMH control the output buffer. When DQML or DQMH is LOW, the corresponding buffer byte is enabled, and when HIGH, disabled. The outputs go to the HIGH impedance state when DQML/DQMH is HIGH. This function corresponds to $\overline{\text{OE}}$ in conventional DRAMs. In write mode, DQML and DQMH control the input buffer. When DQML or DQMH is LOW, the corresponding buffer byte is enabled, and data can be written to the device. When DQML or DQMH is HIGH, input data is masked and cannot be written to the device.
DQ0-DQ7: x8 DQ0-DQ15: x16 DQ0-DQ31: x32	Input/Output	Data on the Data Bus is latched on DQ pins during Write commands, and buffered for output after Read commands.
$\overline{\text{RAS}}$	Input Pin	$\overline{\text{RAS}}$, in conjunction with $\overline{\text{CAS}}$ and $\overline{\text{WE}}$, forms the device command. See the "Command Truth Table" item for details on device commands.
$\overline{\text{WE}}$	Input Pin	$\overline{\text{WE}}$, in conjunction with $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$, forms the device command. See the "Command Truth Table" item for details on device commands.
V _{DDQ}	Power Supply Pin	V _{DDQ} is the output buffer power supply.
V _{DD}	Power Supply Pin	V _{DD} is the device internal power supply.
V _{SSQ}	Power Supply Pin	V _{SSQ} is the output buffer ground.
V _{SS}	Power Supply Pin	V _{SS} is the device internal ground.

COMMAND TRUTH TABLE

Function	CKE		$\overline{CS}$	$\overline{RAS}$	$\overline{CAS}$	$\overline{WE}$	BA1	BA0	A12, A11	
	n - 1	n							A10	A9 - A0
Device deselect (DESL)	H	x	H	x	x	x	x	x	x	x
No operation (NOP)	H	x	L	H	H	H	x	x	x	x
Burst stop (BST)	H	x	L	H	H	L	x	x	x	x
Read	H	x	L	H	L	H	V	V	L	V
Read with auto precharge	H	x	L	H	L	H	V	V	H	V
Write	H	x	L	H	L	L	V	V	L	V
Write with auto precharge	H	x	L	H	L	L	V	V	H	V
Bank activate (ACT)	H	x	L	L	H	H	V	V	V	V
Precharge select bank (PRE)	H	x	L	L	H	L	V	V	L	x
Precharge all banks (PALL)	H	x	L	L	H	L	x	x	H	x
CBR Auto-Refresh (REF)	H	H	L	L	L	H	x	x	x	x
Self-Refresh (SELF)	H	L	L	L	L	H	x	x	x	x
Mode register set (MRS)	H	x	L	L	L	L	L	L	L	V

Note: H=V_{IH}, L=V_{IL} x= V_{IH} or V_{IL}, V = Valid Data.

DQM TRUTH TABLE

Function	CKE		DQMH	DQML
	n-1	n		
Data write / output enable	H	x	L	L
Data mask / output disable	H	x	H	H
Upper byte write enable / output enable	H	x	L	x
Lower byte write enable / output enable	H	x	x	L
Upper byte write inhibit / output disable	H	x	H	x
Lower byte write inhibit / output disable	H	x	x	H

Note:

1. H=V_{IH}, L=V_{IL} x= V_{IH} or V_{IL}, V = Valid Data.
2. x16 options shown. x32 DQM0-DQM3 is similar in function.

IS42/45R86400D/16320D/32160D, IS42/45S86400D/16320D/32160D

AC ELECTRICAL CHARACTERISTICS ^(1,2,3)

Symbol	Parameter		-5		-6		-7		Units
			Min.	Max.	Min.	Max.	Min.	Max.	
t _{CK3}	Clock Cycle Time	CAS Latency = 3	5	—	6	—	7	—	ns
t _{CK2}		CAS Latency = 2	10	—	10	—	7.5	—	ns
t _{AC3}	Access Time From CLK	CAS Latency = 3	—	5.0	—	5.4	—	5.4	ns
t _{AC2}		CAS Latency = 2	—	6	—	6	—	5.4	ns
t _{CH}	CLK HIGH Level Width		2	—	2.5	—	2.5	—	ns
t _{CL}	CLK LOW Level Width		2	—	2.5	—	2.5	—	ns
t _{OH3}	Output Data Hold Time	CAS Latency = 3	2.5	—	2.7	—	2.7	—	ns
t _{OH2}		CAS Latency = 2	2.5	—	2.7	—	2.7	—	ns
t _{LZ}	Output LOW Impedance Time		0	—	0	—	0	—	ns
t _{HZ3}	Output HIGH Impedance Time		—	5	—	5.4	—	5.4	ns
t _{HZ2}			—	6	—	6	—	5.4	ns
t _{DS}	Input Data Setup Time ⁽²⁾		1.8	—	1.5	—	1.5	—	ns
t _{DH}	Input Data Hold Time ⁽²⁾		0.8	—	0.8	—	0.8	—	ns
t _{AS}	Address Setup Time ⁽²⁾		1.5	—	1.5	—	1.5	—	ns
t _{AH}	Address Hold Time ⁽²⁾		0.8	—	0.8	—	0.8	—	ns
t _{CKS}	CKE Setup Time ⁽²⁾		1.5	—	1.5	—	1.5	—	ns
t _{CKH}	CKE Hold Time ⁽²⁾		0.8	—	0.8	—	0.8	—	ns
t _{CMS}	Command Setup Time ($\overline{CS}$, $\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$, $\overline{DQM}$) ⁽²⁾		1.5	—	1.5	—	1.5	—	ns
t _{CMH}	Command Hold Time ($\overline{CS}$, $\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$, $\overline{DQM}$) ⁽²⁾		0.8	—	0.8	—	0.8	—	ns
t _{RC}	Command Period (REF to REF / ACT to ACT)		55	—	60	—	60	—	ns
t _{RAS}	Command Period (ACT to PRE)		38	100K	42	100K	37	100K	ns
t _{RP}	Command Period (PRE to ACT)		15	—	18	—	15	—	ns
t _{RCD}	Active Command To Read / Write Command Delay Time		15	—	18	—	15	—	ns
t _{RRD}	Command Period (ACT [0] to ACT[1])		10	—	12	—	14	—	ns
t _{DPL}	Input Data To Precharge Command Delay time		10	—	12	—	14	—	ns
t _{DAL}	Input Data To Active / Refresh Command Delay time (During Auto-Precharge)		25	—	30	—	29	—	ns
t _{MRD}	Mode Register Program Time		10	—	12	—	14	—	ns
t _{DDE}	Power Down Exit Setup Time		5	—	6	—	7	—	ns
t _{XS}	Self-Refresh Exit Time		60	—	70	—	67	—	ns
t _T	Transition Time		0.3	1.2	0.3	1.2	0.3	1.2	ns
t _{REF}	Refresh Cycle Time (8192)	T _A ≤ 70°C Com, Ind, A1, A2	—	64	—	64	—	64	ms
		T _A ≤ 85°C Ind, A1, A2	—	64	—	64	—	64	ms
		T _A > 85°C A2	—	—	—	—	—	16	ms

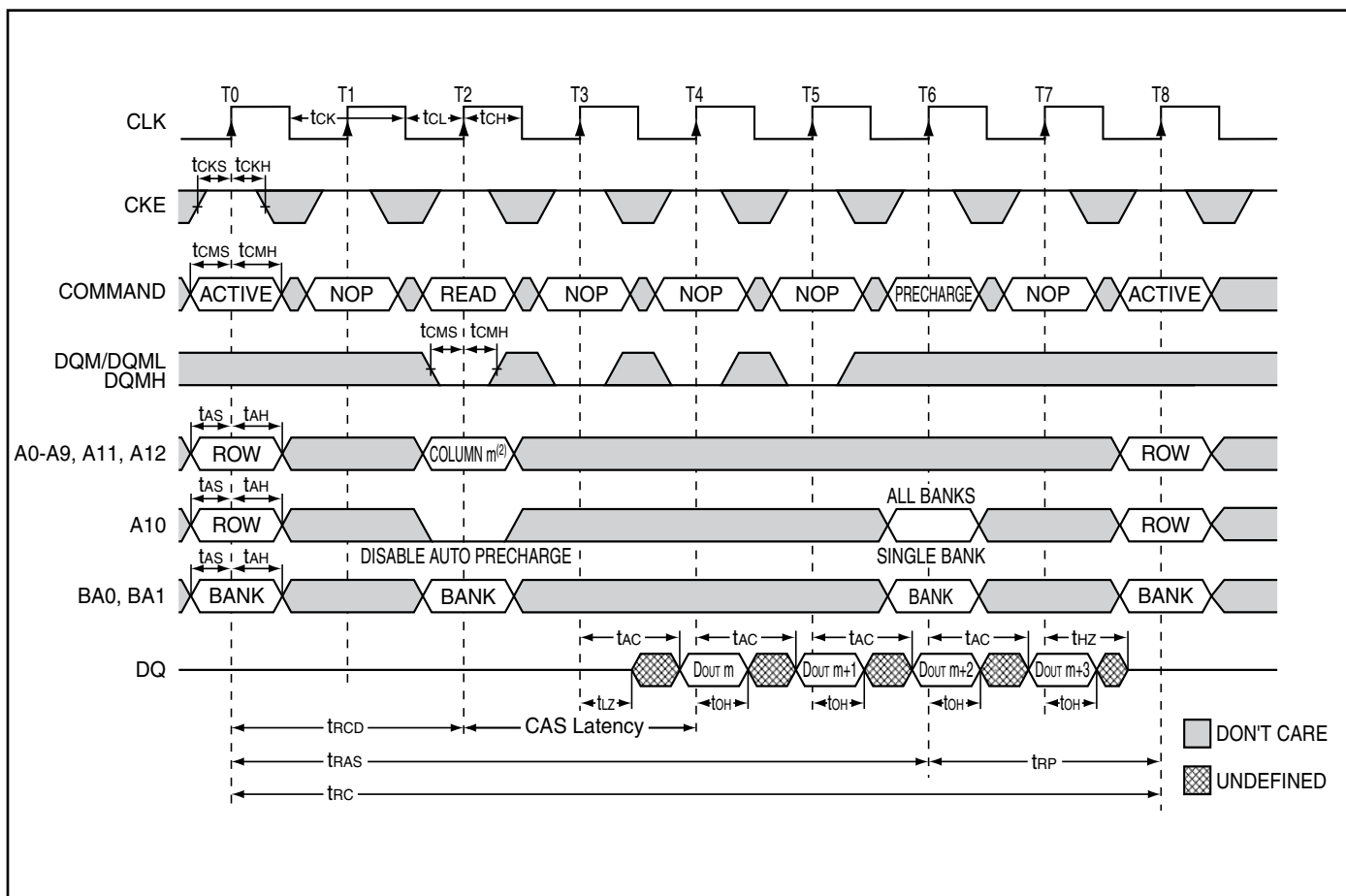
Notes:

1. The power-on sequence must be executed before starting memory operation.
2. Measured with t_r = 1 ns. If clock rising time is longer than 1ns, (t_r / 2 - 0.5) ns should be added to the parameter.
3. The reference level is 1.4V when measuring input signal timing. Rise and fall times are measured between V_{IH}(min.) and V_{IL}(max).

OPERATING FREQUENCY / LATENCY RELATIONSHIPS

SYMBOL	PARAMETER	-5	-6	-7	-7	UNITS
—	Clock Cycle Time	5	6	7	7.5	ns
—	Operating Frequency	200	167	143	133	MHz
t _{CAC}	$\overline{\text{CAS}}$ Latency	3	3	3	2	cycle
t _{RCD}	Active Command To Read/Write Command Delay Time	3	3	3	2	cycle
t _{RAC}	$\overline{\text{RAS}}$ Latency (t _{RCD} + t _{CAC})	$\overline{\text{CAS}}$ Latency = 3	6	6	6	—
		$\overline{\text{CAS}}$ Latency = 2	—	—	—	4
t _{RC}	Command Period (REF to REF / ACT to ACT)	10	10	9	8	cycle
t _{RAS}	Command Period (ACT to PRE)	7	7	6	5	cycle
t _{RP}	Command Period (PRE to ACT)	3	3	3	2	cycle
t _{RRD}	Command Period (ACT[0] to ACT [1])	2	2	2	2	cycle
t _{CCD}	Column Command Delay Time (READ, READA, WRIT, WRITA)	1	1	1	1	cycle
t _{DPL}	Input Data To Precharge Command Delay Time	2	2	2	2	cycle
t _{DAL}	Input Data To Active/Refresh Command Delay Time (During Auto-Precharge)	5	5	5	4	cycle
t _{RBD}	Burst Stop Command To Output in HIGH-Z Delay Time (Read)	$\overline{\text{CAS}}$ Latency = 3	3	3	3	—
		$\overline{\text{CAS}}$ Latency = 2	—	—	—	2
t _{WBD}	Burst Stop Command To Input in Invalid Delay Time (Write)	0	0	0	0	cycle
t _{RQL}	Precharge Command To Output in HIGH-Z Delay Time (Read)	$\overline{\text{CAS}}$ Latency = 3	3	3	3	—
		$\overline{\text{CAS}}$ Latency = 2	—	—	—	2
t _{WDL}	Precharge Command To Input in Invalid Delay Time (Write)	0	0	0	0	cycle
t _{PQL}	Last Output To Auto-Precharge Start Time (Read)	$\overline{\text{CAS}}$ Latency = 3	-2	-2	-2	—
		$\overline{\text{CAS}}$ Latency = 2	—	—	—	-1
t _{QMD}	DQM To Output Delay Time (Read)	2	2	2	2	cycle
t _{DMD}	DQM To Input Delay Time (Write)	0	0	0	0	cycle
t _{MRD}	Mode Register Set To Command Delay Time	2	2	2	2	cycle

READ WITHOUT AUTO PRECHARGE



Notes:

- 1) CAS latency = 2, Burst Length = 4
- 2) x8: A12 = "Don't Care"
- x16: A11, A12 = "Don't Care"
- x32: A9, A11, A12 = "Don't Care"